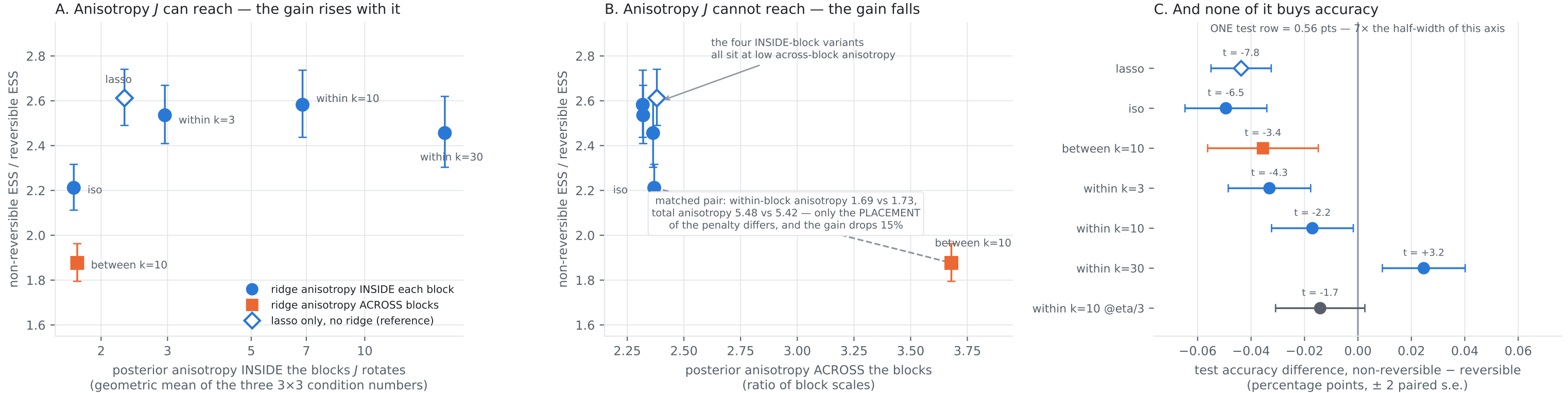


Changing the regularizer moves the non-reversible speed-up from 1.88x to 2.61x, and the accuracy by nothing



TITANIC, ball constraint, exact gradient, elastic net $U = \loglik + w_0^2 / (2 \sigma^2) + \text{lam1} \sum |w_j| + (\text{lam2} / 2) \sum c_j w_j^2$, with U_0 smoothing ONLY the l_1 term -- so $a = \exp(U - U_0)$ depends on $(\text{lam1}, \text{delta})$ alone and is IDENTICAL across every variant shown; the only thing that moves is c . $\text{lam1} = 0.01$ $n_{\text{train}} = 7.12$, $\text{lam2} = 0.05$ $n_{\text{train}} = 35.6$, a in $[0.5, 1]$. $\text{eta} = 1e-4$, $s = (5, 5, 5)$, 96 chains started from a common stationary state and sharing one Gaussian stream, 24000 measured iterations stored every 2, seed 7100 (used nowhere else in the repository). Split-Rhat ≤ 1.031 on every coordinate of every run. ESS from Geyer initial-positive/monotone IACT per chain per coordinate $w_{1..w_9}$; the ratio is the geometric mean over the 9×96 paired chain-coordinates and the bar is ± 2 paired s.e. on the log ratio. `between k=10` and `within k=10` use the SAME multiset of c_j -- the same penalty anisotropy -- placed differently relative to the blocks; their posterior anisotropies are measured, not assumed. Panel C's accuracy is averaged over the whole measured (post-burn-in) run; the axis there spans less than a third of ONE test row out of 179, so several differences are statistically resolvable and none is numerically meaningful. The bottom row re-runs one variant at $\text{eta}/3$ with 3x the iterations, so the simulated time matches: the difference is finite-step bias, not a property of the target, which the two arms provably share.